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Inventor(s)

MOTOKI; Tatsuya et al.

IMAGING APPARATUS

Abstract

An imaging apparatus includes: a housing including a first intake portion, a second intake portion, and an exhaust portion; an air cooling device provided inside the housing with an interval from the exhaust portion to generate an air flow from the first and second intake portions toward the exhaust portion; a rectifying member that is provided at an upstream side of the air flow with respect to the air cooling device and that rectifies intake air from the first intake portion and the second intake portion; a recording unit in which a recording medium to record imaging data is to be disposed; a first heat dissipation member provided at an upstream side of the air flow with respect to the air cooling device and including a heat dissipation fin for dissipating heat of the recording unit; and a heat dissipation sheet that dissipates heat of the recording unit.

Inventors: MOTOKI; Tatsuya (Saitama-shi, JP), SHIMIZU; Takayuki (Saitama-shi, JP), TAKAGI; Masaaki (Saitama-shi, JP)

Applicant: FUJIFILM Corporation (Tokyo, JP)

Family ID: 1000008476262

Assignee: FUJIFILM Corporation (Tokyo, JP)

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS

[0001] The present application claims priority under 35 U.S.C. § 119 to Japanese Patent Application No. 2024-026530 filed on Feb. 26, 2024. The above application is hereby expressly incorporated by reference, in its entirety, into the present application.

BACKGROUND OF THE INVENTION

1. Field of the Invention

[0002] The present invention relates to an imaging apparatus.

2. Description of the Related Art

[0003] JP2020-113889A discloses an imaging apparatus comprising a main body portion that has a first intake hole at a lower right edge portion, a second intake hole at a lower left edge portion, an exhaust hole at an upper left edge portion, and a fan provided near the exhaust hole, an inner space of a first duct and a second duct is formed between the first intake hole and the second intake hole and the fan by an inverted Y-shaped duct separation wall, a heat sink portion for heat dissipation of an IC and a heat sink for heat dissipation of an imaging element are exposed in a first duct in a state of being disposed to be orthogonal to each other, and a socket case in which a medium case including a storage element inside is mounted is exposed in a second duct.

[0004] JP2019-097005A discloses a system camera comprising a heat dissipation module that is provided with a first ventilation port, a second ventilation port, an electric fan, a first exhaust port, and a second exhaust port, in which a heat sink that forms an air flow passage is disposed between the first ventilation port and the second ventilation port and the electric fan, a large number of heat dissipation fins and an inverted Y-shaped rectifying member are provided on the heat sink, and the heat dissipation module is connected to a display module that has a display device such as an LCD to be cooled.

[0005] JP2023-077022A discloses an imaging apparatus in which air sucked from an intake port into an inside of an imaging apparatus main body flows into an inner space of a duct unit formed by a duct base and a duct plate, passes through a plurality of heat dissipation duct fins formed on the duct base, passes through an inside of a cooling fan, and is discharged from an exhaust port, and a first media socket is thermally connected to the duct unit through a second heat conductive sheet, and heat generated in a recording medium is conducted to the duct unit through the first media socket and the second heat conductive sheet.

[0006] JP2022-135869A discloses an imaging apparatus comprising a main frame provided in a housing, in which the main frame supports an imaging unit including an imaging element and a heat sink, an image processing unit including a plurality of IC chips and a heat sink, and a writing unit including a connector into which a recording medium is inserted and a heat sink, a plurality of heat sinks and a fan are disposed in a duct-shaped air flow passage, and an extension length and the number of fins in each heat sink are individually determined based on a heat amount of a unit to be cooled and a surface area of each fin determined by an allowable temperature.

[0007] WO2019/171722A discloses an imaging apparatus in which a flow passage for cooling air is provided in an apparatus main body, the flow passage has a heat dissipation fin disposed on a downstream side of an intake hole, a cooling fan is disposed on a downstream side of the heat dissipation fin, a heat dissipation fin is disposed on a downstream side of the cooling fan, and an exhaust hole is disposed on a downstream side of the heat dissipation fin.

SUMMARY OF THE INVENTION

[0008] One embodiment according to the technology of the present disclosure provides an imaging apparatus capable of improving cooling performance of a recording unit.

(1)

[0009] An imaging apparatus comprising: [0010] a housing that includes a first intake portion, a second intake portion, and an exhaust portion; [0011] an air cooling device that is provided inside the housing with an interval from the exhaust portion and that generates an air flow from the first intake portion and the second intake portion toward the exhaust portion; [0012] a rectifying member that is provided on an upstream side of the air flow with respect to the air cooling device and that rectifies intake air from the first intake portion and the second intake portion; [0013] a recording unit in which a recording medium that records imaging data is disposed; [0014] a first heat dissipation member that is provided on the upstream side of the air flow with respect to the air cooling device and that includes a heat dissipation fin for dissipating heat of the recording unit; and [0015] a second heat dissipation member that is a heat dissipation sheet that dissipates heat of the recording unit.

(2)

[0016] The imaging apparatus according to (1), [0017] in which the recording medium is inserted into and removed from the recording unit.

(3)

[0018] The imaging apparatus according to (1) or (2), [0019] in which the rectifying member is a protruding member.

(4)

[0020] The imaging apparatus according to (3), [0021] in which a height dimension of the protruding member is smaller than a width dimension of the protruding member.

(5)

[0022] The imaging apparatus according to any one of (1) to (4), [0023] in which the recording unit includes a medium accommodation portion in which the recording medium is accommodated, and a first electronic member that writes the imaging data to the recording medium accommodated in the medium accommodation portion, [0024] the first heat dissipation member dissipates heat of the first electronic member, and [0025] the second heat dissipation member dissipates heat of the medium accommodation portion.

(6)

[0026] The imaging apparatus according to (5), [0027] in which the heat dissipated from the medium accommodation portion includes heat generated from the recording medium.

(7)

[0028] The imaging apparatus according to (5) or (6), [0029] in which heat conduction occurs between the medium accommodation portion and the recording medium.

(8)

[0030] The imaging apparatus according to any one of (1) to (7), further comprising: [0031] a second electronic member that is different from the recording unit; and [0032] a third heat dissipation member that is provided on the upstream side of the air flow with respect to the air cooling device and that includes a heat dissipation fin that dissipates heat of the second electronic member, [0033] in which an arrangement direction of fins in the first heat dissipation member and an arrangement direction of fins in the third heat dissipation member are different from each other.

(9)

[0034] The imaging apparatus according to (8), [0035] in which the first heat dissipation member and the third heat dissipation member are disposed in the same space.

(10)

[0036] The imaging apparatus according to any one of (1) to (9), comprising: [0037] an imaging unit that includes an imaging element and a third electronic member; and [0038] a fourth heat dissipation member that includes a heat dissipation fin for dissipating heat of the third electronic

member, [0039] in which the fourth heat dissipation member includes a first heat dissipation fin and a second heat dissipation fin, and [0040] at least a part of the air cooling device is provided between the first heat dissipation fin and the second heat dissipation fin.

(11)

[0041] The imaging apparatus according to any one of (1) to (10), comprising: [0042] a second electronic member that is different from the recording unit; [0043] a third heat dissipation member that is provided on the upstream side of the air flow with respect to the air cooling device and that includes a heat dissipation fin that dissipates heat of the second electronic member; [0044] an imaging unit that includes an imaging element and a third electronic member; and [0045] a fourth heat dissipation member that includes a heat dissipation fin for dissipating heat of the third electronic member, [0046] in which an arrangement interval of fins in the first heat dissipation member, an arrangement interval of fins in the third heat dissipation member, and an arrangement interval of fins in the fourth heat dissipation member are different from each other.

(12)

[0047] The imaging apparatus according to any one of (1) to (11), [0048] in which the air cooling device is an air cooling fan.

(13)

[0049] The imaging apparatus according to any one of (1) to (12), [0050] in which the second heat dissipation member is a metal sheet or a graphite sheet.

(14)

[0051] The imaging apparatus according to any one of (1) to (13), [0052] in which the recording unit includes a medium accommodation portion in which the recording medium is accommodated and that has a lid portion, [0053] the lid portion includes a fixing member that dissipates heat of the medium accommodation portion, and [0054] the second heat dissipation member is connected to the fixing member.

(15)

[0055] The imaging apparatus according to (14), [0056] in which the second heat dissipation member is connected to the first heat dissipation member.

(16)

[0057] An imaging apparatus comprising: [0058] a housing that includes a first intake portion, a second intake portion, and an exhaust portion; [0059] an air cooling device that is provided inside the housing with an interval from the exhaust portion and that generates an air flow from the first intake portion and the second intake portion toward the exhaust portion; [0060] a rectifying member that is provided on an upstream side of the air flow with respect to the air cooling device and that rectifies intake air from the first intake portion and the second intake portion; [0061] a recording unit in which a recording medium that records imaging data is inserted into and removed from; and [0062] a second heat dissipation member that is a heat dissipation sheet that dissipates heat of the recording unit.

[0063] According to the present invention, it is possible to provide an imaging apparatus capable of improving cooling performance of a recording unit.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0064] FIG. 1 is a simplified perspective view showing an example of an imaging apparatus **100** in the present embodiment as viewed from the front left side.

[0065] FIG. 2 is a simplified perspective view of the imaging apparatus **100** shown in FIG. 1 as viewed from the front right side.

[0066] FIG. 3 is a diagram showing an example of a flow of air in a housing **110**.

[0067] FIG. 4 is a view of a heat dissipation member provided in the housing **110** as viewed from a rear side of the imaging apparatus **100**.

[0068] FIG. 5 is a view of a heat dissipation member provided in the housing **110** as viewed from a left side of the imaging apparatus **100**.

[0069] FIG. 6 is a perspective view showing an example of a disposition relationship between a fourth heat dissipation member **118** and an air cooling fan **111**.

[0070] FIG. 7 is a perspective view showing an example of a disposition relationship between a first heat dissipation member **113** and a third heat dissipation member **116**.

[0071] FIG. 8 is a diagram showing a first modification example of a configuration of the first heat dissipation member **113**.

[0072] FIG. 9 is a diagram showing a second modification example of the configuration of the first heat dissipation member **113**.

[0073] FIG. 10 is a diagram showing a first attachment example of the second heat dissipation member **114**.

[0074] FIG. 11 is a diagram showing a second attachment example of the second heat dissipation member **114**.

[0075] FIG. 12 is a diagram showing a third attachment example of the second heat dissipation member **114**.

[0076] FIG. 13 is a diagram showing a fourth attachment example of the second heat dissipation member **114**.

[0077] FIG. 14 is a diagram showing a state where the lid portion **119** shown in FIG. 13 is opened.

DESCRIPTION OF THE PREFERRED EMBODIMENTS

[0078] Hereinafter, an example of an embodiment of the present invention will be described with reference to the drawings. In the present embodiment, the terms “upward direction”, “downward direction”, “leftward direction”, “rightward direction”, “forward direction”, and “backward direction” are used, but these directions are relative directions set for convenience of description for the imaging apparatus shown in each drawing.

Imaging Apparatus of Embodiment

[0079] FIG. 1 is a simplified perspective view showing an example of the imaging apparatus **100** in the present embodiment as viewed from the front left side. FIG. 2 is a simplified perspective view of the imaging apparatus **100** shown in FIG. 1 as viewed from the front right side. Examples of the imaging apparatus include a digital camera or a single-lens reflex camera capable of capturing a motion picture. As shown in FIGS. 1 and 2, the imaging apparatus **100** comprises a housing **110** that constitutes a main body portion of the imaging apparatus **100**, and a lens **120** attached to the housing **110**.

[0080] The housing **110** is formed in a substantially cubic shape. A first intake portion **110a** is provided at a lower left end portion of the housing **110**. In addition, a second intake portion **110b** is provided at a lower right end portion of the housing **110**. The first intake portion **110a** and the second intake portion **110b** are opening portions capable of taking in air from the outside of the housing **110** into the inside of the housing **110**. The first intake portion **110a** and the second intake portion **110b** are composed of, for example, a plurality of slit-shaped openings.

[0081] In addition, an exhaust portion **110c** is provided at an upper right end portion of the housing **110**. The exhaust portion **110c** is an opening portion through which air inside the housing **110** can be discharged to the outside of the housing **110**. The exhaust portion **110c** is composed of, for example, a plurality of slit-shaped openings.

[0082] The opening constituting the exhaust portion **110c** is formed, for example, from an upper wall of the housing **110** to a right side wall of the housing **110**. Although not shown, the opening constituting the first intake portion **110a** is formed from the left side wall to the lower wall, and the opening constituting the second intake portion **110b** is formed from the right side wall to the lower wall.

[0083] In addition, a lid portion **119** of a recording unit **112** (described below with reference to FIG. 5) into which a recording medium is inserted into and removed from is provided on a lower portion of the left side wall of the housing **110**. The recording medium records the imaging data of the imaging apparatus **100**.

[0084] The lens **120** is attached to a front side (front surface) of the housing **110**. The lens **120** can be replaced. The lens **120** includes at least an imaging lens such as a focus lens or a zoom lens.

[0085] FIG. 3 is a diagram showing an example of a flow of air in the housing **110**. In FIG. 3, components unnecessary for the description are omitted in order to make the flow of air easy to see. As shown in FIG. 3, an air cooling fan **111**, which is an example of an “air cooling device” according to the embodiment of the present invention, is provided in a substantially central portion of a housing **110**. The air cooling fan **111** is provided with an interval from an exhaust portion **110c** provided at an upper right end portion of the housing **110**. The description that the air cooling fan **111** is provided with an interval from an exhaust portion **110c** means that the air cooling fan **111** is not provided in the vicinity of the exhaust portion **110c**. The expression “provided with an interval” means, for example, that there is a space between the air cooling fan **111** and the exhaust portion **110c**, in which a heat generating body such as an electronic component can be disposed, or there is a space in which the heat generating body is actually disposed. The air cooling device may be, for example, a blower or a compressor.

[0086] The air cooling fan **111** generates an air flow from the first intake portion **110a** and the second intake portion **110b** toward the exhaust portion **110c** in the housing **110**. For example, the air cooling fan **111** rotates to allow air outside the housing **110** to flow into the housing **110** from the first intake portion **110a** provided at the lower left end portion of the housing **110** and the second intake portion **110b** provided at the lower right end portion of the housing **110**, to pass through the air cooling fan **111**, and to flow out of the housing **110** from the exhaust portion **110c** provided at the upper right end portion of the housing **110**, as indicated by the solid line arrow in FIG. 3.

[0087] In addition, a rectifying member **110d** for rectifying a flow of air in the housing **110** is provided at a central lower end portion in the housing **110**. The rectifying member **110d** is provided on the upstream side (the lower side in FIG. 3) of the air flow with respect to the air cooling fan **111** indicated by the solid line arrow in FIG. 3. The upstream side of the air flow refers to the first intake portion **110a** and the second intake portion **110b** side in the housing **110** (not the exhaust portion **110c** side). The rectifying member **110d** is a protruding member that protrudes from the lower wall of the housing **110** toward the upper wall direction. The height dimension H of the rectifying member **110d** is smaller than the width dimension W. The rectifying member **110d** rectifies the intake air from the first intake portion **110a** and the second intake portion **110b**. The rectifying member **110d** rectifies the air flowing into the housing **110** from the first intake portion **110a** and the second intake portion **110b** to rise in the direction of the air cooling fan **111** as indicated by a solid line arrow.

[0088] FIG. 4 is a view of a heat dissipation member provided in the housing **110** as viewed from a rear side of the imaging apparatus **100**. FIG. 5 is a view of the heat dissipation member provided in the housing **110** as viewed from the left side of the imaging apparatus **100**. As shown in FIGS. 4 and 5, a first heat dissipation member **113**, a second heat dissipation member **114**, a third heat dissipation member **116**, and a fourth heat dissipation member **118** are provided in the housing **110**. In addition, a recording unit **112**, a main board **115**, and an imaging unit **117** are provided in the housing **110**.

[0089] The first heat dissipation member **113** and the second heat dissipation member **114** are heat dissipation members attached to the recording unit **112**. The first heat dissipation member **113** is composed of a heat sink having a plurality of heat dissipation fins. The first heat dissipation member **113** is provided on the upstream side (the lower side in FIGS. 4 and 5) of the air flow indicated by the solid line arrow in FIG. 3 with respect to the air cooling fan **111**. The second heat

dissipation member **114** is composed of a heat transfer sheet or a heat dissipation sheet having high heat transfer properties. The second heat dissipation member is, for example, a metal sheet, a graphite sheet, a copper foil tape, and an aluminum tape that can be attached to the recording unit **112**.

[0090] The recording unit **112** is a device into which a recording medium for recording the imaging data of the imaging apparatus **100** is inserted in and removed from. The recording unit **112** is provided at a lower left portion of the rear portion in the housing **110**. In the present example, the recording unit **112** is provided on the rear side of the air cooling fan **111** along the rear wall of the housing **110**. The recording unit **112** includes a medium accommodation portion **112a** and a first electronic member **112b**.

[0091] The medium accommodation portion **112a** is a portion in which the recording medium is accommodated. The medium accommodation portion **112a** is provided with an insertion port **112c** for inserting in and pulling out a recording medium. The medium accommodation portion **112a** is disposed such that the insertion port **112c** is exposed to the outside from the left side wall of the housing **110**. The lid portion **119** shown in FIG. 1 is attached to the insertion port **112c** of the medium accommodation portion **112a**. The first electronic member **112b** is, for example, an electronic board that performs processing of writing imaging data to the recording medium accommodated in the medium accommodation portion **112a**. The recording medium is, for example, a CFexpress Type B or XQD memory card.

[0092] The first heat dissipation member **113** and the second heat dissipation member **114** absorb heat emitted from the recording unit **112** and dissipate the heat into the air to cool the recording unit **112**. The first heat dissipation member **113** is attached to the first electronic member **112b** of the recording unit **112** and dissipates heat of the first electronic member **112b**. The second heat dissipation member **114** is attached to the medium accommodation portion **112a** of the recording unit **112** and dissipates heat of the medium accommodation portion **112a**. Heat conduction occurs between the medium accommodation portion **112a** and the recording medium accommodated therein. Therefore, the heat generated by the recording medium is included in the heat dissipated from the medium accommodation portion **112a**. The second heat dissipation member **114** dissipates heat of the medium accommodation portion **112a** and the recording medium accommodated therein.

[0093] The third heat dissipation member **116** is a heat dissipation member attached to the main board **115** which is an example of a “second electronic member” according to the embodiment of the present invention, and is configured with a heat sink having a plurality of heat dissipation fins. The third heat dissipation member **116** is provided on the upstream side (the lower side in FIGS. 4 and 5) of the air flow with respect to the air cooling fan **111** indicated by the solid line arrow in FIG. 3. The third heat dissipation member **116** is disposed in the same space as the first heat dissipation member **113** on the upstream side of the air flow with respect to the air cooling fan **111**.

[0094] The main board **115** is, for example, an electronic board that performs imaging control, image processing, and the like of the imaging apparatus **100**. The main board **115** is provided at a right end portion in the housing **110**. In the present example, the main board **115** is provided on the right side of the air cooling fan **111** along the right side wall of the housing **110**. The third heat dissipation member **116** absorbs heat emitted from the main board **115** and dissipates the heat into the air to cool the main board **115**.

[0095] The fourth heat dissipation member **118** is a heat dissipation member attached to the imaging unit **117**, and is composed of a heat sink having a plurality of heat dissipation fins. A part of the fourth heat dissipation member **118** is provided on the upstream side (lower side in FIGS. 4 and 5) of the air flow with respect to the air cooling fan **111** indicated by the solid line arrow in FIG. 3.

[0096] The imaging unit **117** includes an imaging element **117a** and a third electronic member **117b**. The imaging unit **117** is provided at a front portion in the housing **110** to face the lens **120**. In

the present example, the imaging unit **117** is provided on the front side of the air cooling fan **111** along the front wall of the housing **110**. The imaging element **117a** and the third electronic member **117b** are configured such that the third electronic member **117b** is disposed on a rear surface side of the imaging element **117a** in an optical axis direction of the lens **120**. The light incident from the lens **120** is imaged on the imaging element **117a**.

[0097] The imaging element **117a** is composed of, for example, a charge coupled device (CCD) image sensor or a Complementary Metal Oxide Semiconductor (CMOS) image sensor. The third electronic member **117b** includes a signal processing circuit that performs sampling processing, digital conversion processing, and the like on the captured image signal output from the imaging element **117a**. In addition, the third electronic member **117b** includes an image processing unit that performs digital signal processing on the captured image signal after processing by the signal processing circuit to generate captured image data, for example, in a joint photographic experts group (JPEG) format.

[0098] FIG. **6** is a perspective view showing an example of a disposition relationship between the fourth heat dissipation member **118** and the air cooling fan **111**. As shown in FIG. **6**, the fourth heat dissipation member **118** has a base portion **118a**, a first heat dissipation fin **118b**, and a second heat dissipation fin **118c**.

[0099] The base portion **118a** is a plate-shaped member to which the third electronic member **117b** is attached. The third electronic member **117b** is attached to a surface on a side on which the lens **120** is provided (surface of the front side), of the base portion **118a** disposed in the housing **110**. The base portion **118a** is made of a metal member such as aluminum or copper having high thermal conductivity.

[0100] The first heat dissipation fin **118b** and the second heat dissipation fin **118c** are provided on a surface of the base portion **118a** on a side opposite to a side to which the third electronic member **117b** is attached (surface on the rear side). The first heat dissipation fin **118b** is a heat dissipation fin provided on an upper portion of a surface of the base portion **118a** on a rear side, and is composed of a plurality of fins extending in parallel to each other from the base portion **118a**. The second heat dissipation fin **118c** is a heat dissipation fin provided on a lower portion of the surface of the base portion **118a** on the rear side, and is composed of a plurality of fins extending in parallel to each other from the base portion **118a**, similarly to the first heat dissipation fin **118b**. The plurality of fins of the first heat dissipation fin **118b** and the second heat dissipation fin **118c** are arranged in the left-right direction.

[0101] The first heat dissipation fin **118b** and the second heat dissipation fin **118c** are fins that dissipate heat of the third electronic member **117b**. The fins for dissipating heat of the third electronic member **117b** are heat conduction members in which the first heat dissipation fin **118b** and the second heat dissipation fin **118c** are thermally connected to the third electronic member **117b**. For example, the first heat dissipation fin **118b** and the second heat dissipation fin **118c** may be in direct contact with the third electronic member **117b**, or may be connected to the third electronic member **117b** via another heat conduction member. The first heat dissipation fin **118b** and the second heat dissipation fin **118c** of the present example are connected to the third electronic member **117b** via the base portion **118a**. The first heat dissipation fin **118b** and the second heat dissipation fin **118c** dissipate the heat conducted from the third electronic member **117b** to the base portion **118a**.

[0102] At least a part of the air cooling fan **111** is provided between the first heat dissipation fin **118b** and the second heat dissipation fin **118c**. The description that “at least a part of the air cooling fan **111** is provided between the first heat dissipation fin **118b** and the second heat dissipation fin **118c**” means that the first heat dissipation fin **118b** and the second heat dissipation fin **118c** are provided in a blowing direction of the air cooling fan **111**. In the present example, a part of the air cooling fan **111** on the front side is provided between the first heat dissipation fin **118b** and the second heat dissipation fin **118c**. With such disposition, the first heat dissipation fin **118b** and the

second heat dissipation fin **118c** can discharge the heat of the third electronic member **117b** into the air flow of the air cooling fan **111** (in the air flow indicated by the solid line arrow in FIG. 3). [0103] FIG. 7 is a perspective view showing an example of a disposition relationship between the first heat dissipation member **113** and the third heat dissipation member **116**. As shown in FIG. 7, the first heat dissipation member **113** has a base portion **113a** and a heat dissipation fin **113b**. In addition, the third heat dissipation member **116** has a base portion **116a** and a heat dissipation fin **116b**.

[0104] A base portion **113a** of the first heat dissipation member **113** is a plate-shaped member to which the first electronic member **112b** of the recording unit **112** is attached. The first electronic member **112b** is attached to a surface on a side on which the rear wall of the housing **110** is provided (surface on the rear side), of the base portion **113a** disposed in the housing **110**. The base portion **113a** is made of a metal member such as aluminum or copper having high thermal conductivity.

[0105] The heat dissipation fin **113b** is provided on a surface on a side opposite to a side on which the first electronic member **112b** is attached (surface on the front side) of the base portion **113a**, and are composed of a plurality of fins extending in parallel to each other from the base portion **113a**. The heat dissipation fin **113b** is a fin that dissipates heat of the first electronic member **112b**. The fin for dissipating heat of the first electronic member **112b** refers to a heat conduction member in which the heat dissipation fin **113b** is thermally connected to the first electronic member **112b**. For example, the heat dissipation fin **113b** may be in direct contact with the first electronic member **112b** or may be connected to the first electronic member **112b** via another heat conduction member. The heat dissipation fin **113b** of the present example is connected to the first electronic member **112b** via the base portion **113a**. The heat dissipation fin **113b** dissipates heat conducted from the first electronic member **112b** to the base portion **113a**. As described in FIGS. 4 and 5, the first heat dissipation member **113** is provided on the upstream side of the air flow with respect to the air cooling fan **111**. However, the heat dissipation fin **113b** of the first heat dissipation member **113** only need to be provided on the upstream side of the air flow with respect to the air cooling fan **111**, and the base portion **113a** may extend to the height of the air cooling fan **111**.

[0106] A base portion **116a** of the third heat dissipation member **116** is a plate-shaped member to which the main board **115** is attached. The main board **115** is attached to a surface on a side on which the right wall of the housing **110** is provided (surface on the right side), of the base portion **116a** disposed in the housing **110**. The base portion **116a** is made of a metal member such as aluminum or copper having high thermal conductivity.

[0107] The heat dissipation fins **116b** are provided on a surface on a side opposite to a side of the base portion **116a** to which the main board **115** is attached (surface on the left side) and are composed of a plurality of fins extending in parallel to each other from the base portion **116a**. The heat dissipation fin **116b** is a fin that dissipates heat of the main board **115**. The fin that dissipates heat of the main board **115** is heat conduction member in which the heat dissipation fin **116b** is thermally connected to the main board **115**. For example, the heat dissipation fin **116b** may be in direct contact with the main board **115** or may be connected to the main board **115** via another heat conduction member. The heat dissipation fin **116b** of the present example is connected to the main board **115** via the base portion **116a**. The heat dissipation fin **116b** dissipates heat conducted from the main board **115** to the base portion **116a**.

[0108] An arrangement direction of the heat dissipation fins **113b** in the first heat dissipation member **113** is different from an arrangement direction of the heat dissipation fins **116b** in the third heat dissipation member **116**. The arrangement direction of the heat dissipation fins **113b** and the arrangement direction of the heat dissipation fins **116b** are different from each other by, for example, about 90°. The plurality of fins in the heat dissipation fins **113b** are arranged in the left-right direction. On the other hand, the plurality of fins in the heat dissipation fins **116b** are arranged in the front-rear direction.

[0109] The arrangement interval of the heat dissipation fins **113b** in the first heat dissipation member **113**, the arrangement interval of the heat dissipation fins **116b** in the third heat dissipation member **116**, and the arrangement intervals of the first heat dissipation fins **118b** and the second heat dissipation fins **118c** (see FIG. 6) in the fourth heat dissipation member **118** are different from each other.

[0110] The arrangement direction of the heat dissipation fins **113b** in the first heat dissipation member **113** and the arrangement direction of the first heat dissipation fins **118b** and the second heat dissipation fins **118c** in the fourth heat dissipation member **118** are the same left-right direction. However, the arrangement interval of the heat dissipation fins **113b** and the arrangement interval of the first heat dissipation fins **118b** and the second heat dissipation fins **118c** are different from each other. Therefore, it is difficult to dispose the heat dissipation fins to be alternated so that the heat dissipation fins of both the heat dissipation members do not interfere with each other. Therefore, the cutout portion **113c** is formed in the heat dissipation fins **113b** of the first heat dissipation member **113** so that the heat dissipation fins do not interfere with each other regardless of the difference in the arrangement interval between the two heat dissipation fins.

[0111] As described above, the imaging apparatus **100** of the present embodiment comprises the first heat dissipation member **113** that includes the heat dissipation fin **113b** that is provided on the upstream side of the air flow with respect to the air cooling fan **111** and that dissipates heat of the first electronic member **112b** in the recording unit **112**, and the second heat dissipation member **114** consisting of the heat dissipation sheet that dissipates heat of the medium accommodation portion **112a** in the recording unit **112**. According to this configuration, the heat generated in the first electronic member **112b** of the recording unit **112** is conducted to the first heat dissipation member **113**, and the heat generated in the medium accommodation portion **112a** of the recording unit **112** is conducted to the second heat dissipation member **114**, and the first heat dissipation member **113** and the second heat dissipation member **114** to which the heat has been conducted can be effectively cooled by a low temperature air flow generated by the air cooling fan **111** using the outside air suctioned from the first intake portion **110a** and the second intake portion **110b**. In this way, by performing the heat dissipation of the recording unit **112** using the first heat dissipation member **113** and the second heat dissipation member **114**, the cooling property of the recording unit **112** can be improved.

[0112] In addition, the imaging apparatus **100** further comprises a third heat dissipation member **116** that is provided on an upstream side of the air flow with respect to the air cooling fan **111** and includes a heat dissipation fin **116b** that dissipates heat of the main board **115**, and an arrangement direction of the heat dissipation fins **113b** of the first heat dissipation member **113** dissipating heat of the recording unit **112** and an arrangement direction of the heat dissipation fins **116b** of the third heat dissipation member **116** dissipating heat of the main board **115** are different from each other. According to this configuration, it is possible to suppress interference between the heat dissipation from the heat dissipation fin **113b** in the first heat dissipation member **113** and the heat dissipation from the heat dissipation fin **116b** in the third heat dissipation member **116**. As a result, the cooling property of the recording unit **112** can be further improved.

[0113] In addition, the imaging apparatus **100** further comprises a fourth heat dissipation member **118** that includes a first heat dissipation fin **118b** and a second heat dissipation fin **118c** that dissipate heat of the third electronic member **117b** in the imaging unit **117**, and at least a part of the air cooling fan **111** is provided between the first heat dissipation fin **118b** and the second heat dissipation fin **118c**. According to this configuration, it is possible to effectively use the space in the housing **110**, and to reduce the size of the imaging apparatus **100**.

[0114] In addition, in the imaging apparatus **100**, the arrangement interval of the heat dissipation fins **113b** in the first heat dissipation member **113**, the arrangement interval of the heat dissipation fins **116b** in the third heat dissipation member **116**, and the arrangement interval of the first heat dissipation fins **118b** and the second heat dissipation fins **118c** in the fourth heat dissipation

member **118** are different from each other. According to this configuration, the cooling property of the recording unit **112** having a large amount of heat dissipation can be further improved by changing the arrangement interval of the heat dissipation fins in accordance with the amount of heat dissipated by the electronic member attached to each heat dissipation member and the flow passage of the cooling air. In addition, in the imaging apparatus **100**, since the cutout portion **113c** is provided in the heat dissipation fin **113b** of the first heat dissipation member **113**, the interference between the heat dissipation fin **113b** and the first heat dissipation fin **118b** and the second heat dissipation fin **118c** of the fourth heat dissipation member **118** can be suppressed, and the heat dissipation property of the recording unit **112** by the first heat dissipation member **113** can be maintained.

Modification Example of First Heat Dissipation Member **113**

[0115] FIG. **8** is a diagram showing a first modification example of the configuration of the first heat dissipation member **113**. As shown in FIG. **8**, the heat dissipation fin **113b** of the first heat dissipation member **113** may be provided with a hole **113d** in each heat dissipation fin **113b**. In the first modification example, a plurality of circular holes are provided as the hole **113d**. The shape and the number of the hole are not particularly limited. In addition, the present invention is not limited to the first heat dissipation member **113**, and the heat dissipation fins of other heat dissipation members (the third heat dissipation member **116** and the fourth heat dissipation member **118**) may also be provided with holes in the same manner.

[0116] FIG. **9** is a diagram showing a second modification example of the configuration of the first heat dissipation member **113**. As shown in FIG. **9**, a cut and raised portion **113e** may be provided in each of the heat dissipation fins **113b** of the first heat dissipation member **113**. The cut and raised portion **113e** is formed by processing of cutting the heat dissipation fin **113b** by punching or the like and bending the cut portion to form the cut portion into, for example, an L-shape. The cut and raised portion may be provided in the heat dissipation fins of the third heat dissipation member **116** and the fourth heat dissipation member **118** in the same manner.

[0117] As described above, by providing the hole **113d** or the cut and raised portion **113e** in the heat dissipation fin **113b** of the first heat dissipation member **113**, it is possible to prevent the flow of heat generated by the first electronic member **112b** (recording unit **112**) from stagnating between the heat dissipation fins **113b**, and it is possible to improve the heat dissipation effect.

Attachment Example of Second Heat Dissipation Member **114**

[0118] FIG. **10** is a view showing a first attachment example of the second heat dissipation member **114**. As shown in FIG. **10**, a graphite sheet, which is an example of the second heat dissipation member **114**, is attached over a left wall **110e** of the housing **110** and the medium accommodation portion **112a**. Specifically, the second heat dissipation member **114** is attached over the inner side surface of the left wall **110e** to the upper surface of the medium accommodation portion **112a**. As a result, heat generated in the medium accommodation portion **112a** can be conducted to the wall **110e** of the housing **110** through the graphite sheet, and the heat dissipation effect can be improved.

[0119] FIG. **11** is a diagram showing a second attachment example of the second heat dissipation member **114**. As shown in FIG. **11**, a plurality of second heat dissipation members **114** (for example, graphite sheets) may be stacked and attached. For example, another second heat dissipation member **114b** may be attached between the second heat dissipation member **114a** attached over the left wall **110e** of the housing **110** and the medium accommodation portion **112a**, and the medium accommodation portion **112a**.

[0120] In addition, the second heat dissipation member **114b**, which is stacked and attached, may be extended from the medium accommodation portion **112a**, and the extended portion may be attached to another portion of the housing **110** or the first electronic member **112b** of the recording unit **112**. In the case of the present example, the second heat dissipation member **114b**, which is stacked and attached, is attached over the medium accommodation portion **112a** and the first electronic member **112b**. In the present example, two graphite sheets are attached. However the

present invention is not limited thereto. For example, one graphite sheet may be attached over the housing **110**, the medium accommodation portion **112a**, and the first electronic member **112b**. According to the second attachment example, the heat generated in the medium accommodation portion **112a** can be conducted through the graphite sheet, not only to the housing **110**, but also to the first heat dissipation member **113** attached to the first electronic member **112b** through the graphite sheet and the first electronic member **112b**, and thus the heat dissipation effect can be further improved.

[0121] FIG. **12** is a diagram showing a third attachment example of the second heat dissipation member **114**. As shown in FIG. **12**, the second heat dissipation member **114c**, which is stacked and attached on the second heat dissipation member **114a** (for example, the graphite sheet), may be extended to the lower side of the medium accommodation portion **112a** so as to be attached over the first heat dissipation member **113** to which the first electronic member **112b** is attached.

According to the third attachment example, the heat generated in the medium accommodation portion **112a** can be conducted through the graphite sheet not only to the housing **110**, but also to the first heat dissipation member **113**, and thus the heat dissipation effect can be further improved.

[0122] FIG. **13** is a diagram showing a fourth attachment example of the second heat dissipation member **114**. As shown in FIG. **13**, the second heat dissipation member **114** (for example, a graphite sheet) may be attached to the lid portion **119** of the medium accommodation portion **112a**.

[0123] The lid portion **119** rotates around the rotary shaft **119c** so as to be attached to be openable and closable with respect to the left wall **110e** of the housing **110**. The lid portion **119** of FIG. **13** shows a closed state. The lid portion **119** includes a fixing member **119a** that dissipates heat of the medium accommodation portion **112a** and a resin cover **119b** that is provided in a central portion of the fixing member **119a**. The fixing member **119a** is composed of, for example, a metal member such as aluminum or copper having high thermal conductivity.

[0124] The second heat dissipation member **114** is attached over the left wall **110e** of the housing **110** and the lid portion **119**. Specifically, the second heat dissipation member **114** is attached over the outer side surface of the left wall **110e**, the outer side surface of the fixing member **119a**, and the inner side surface of the resin cover **119b**. The second heat dissipation member **114** is connected to the fixing member **119a**.

[0125] FIG. **14** is a view showing a state in which the lid portion **119** shown in FIG. **13** is opened. As shown in FIG. **14**, by opening the lid portion **119**, the insertion port **112c** of the medium accommodation portion **112a** is exposed from the left wall **110e** of the housing **110**, and the recording medium can be inserted in and removed from. According to the third attachment example, the heat generated in the medium accommodation portion **112a** can be conducted to the wall **110e** of the housing **110** via the graphite sheet attached to the lid portion **119**, and the heat dissipation effect can be improved.

EXPLANATION OF REFERENCES

[0126] **100**: imaging apparatus [0127] **110**: housing [0128] **110a**: first intake portion [0129] **110b**: second intake portion [0130] **110c**: exhaust portion [0131] **110d**: rectifying member [0132] **110e**: wall [0133] **111**: air cooling fan [0134] **112**: recording unit [0135] **112a**: medium accommodation portion [0136] **112b**: first electronic member [0137] **112c**: insertion port [0138] **113**: first heat dissipation member [0139] **113a**, **116a**, **118a**: base portion [0140] **113b**, **116b**: heat dissipation fin [0141] **113c**: cutout portion [0142] **113d**: hole [0143] **113e**: cut and raised portion [0144] **114**, **114a**, **114b**, **114c**: second heat dissipation member [0145] **115**: main board [0146] **116**: third heat dissipation member [0147] **117**: imaging unit [0148] **117a**: imaging element [0149] **117b**: third electronic member [0150] **118**: fourth heat dissipation member [0151] **118b**: first heat dissipation fin [0152] **118c**: second heat dissipation fin [0153] **119**: lid portion [0154] **119a**: fixing member [0155] **119b**: resin cover [0156] **119c**: rotation shaft [0157] **120**: lens

Claims

1. An imaging apparatus comprising: a housing that includes a first intake portion, a second intake portion, and an exhaust portion; an air cooling device that is provided inside the housing with an interval from the exhaust portion and that is configured to generate an air flow from the first intake portion and the second intake portion toward the exhaust portion; a rectifying member that is provided at an upstream side of the air flow with respect to the air cooling device and that rectifies intake air from the first intake portion and the second intake portion; a recording unit in which a recording medium that is to record imaging data is to be disposed; a first heat dissipation member that is provided at an upstream side of the air flow with respect to the air cooling device and that includes a heat dissipation fin for dissipating heat of the recording unit; and a second heat dissipation member that is a heat dissipation sheet that dissipates heat of the recording unit.
2. The imaging apparatus according to claim 1, wherein the recording medium is capable of being inserted into and removed from the recording unit.
3. The imaging apparatus according to claim 1, wherein the rectifying member is a protruding member.
4. The imaging apparatus according to claim 3, wherein a height dimension of the protruding member is smaller than a width dimension of the protruding member.
5. The imaging apparatus according to claim 1, wherein the recording unit comprises a medium accommodation portion in which the recording medium is to be accommodated, and a first electronic member that is configured to write the imaging data to the recording medium accommodated in the medium accommodation portion, the first heat dissipation member dissipates heat of the first electronic member, and the second heat dissipation member dissipates heat of the medium accommodation portion.
6. The imaging apparatus according to claim 5, wherein the heat dissipated from the medium accommodation portion includes heat generated from the recording medium.
7. The imaging apparatus according to claim 5, wherein heat conduction occurs between the medium accommodation portion and the recording medium.
8. The imaging apparatus according to claim 1, further comprising: a second electronic member that is different from the recording unit; and a third heat dissipation member that is provided at an upstream side of the air flow with respect to the air cooling device and that includes a heat dissipation fin that dissipates heat of the second electronic member, wherein an arrangement direction of the fin in the first heat dissipation member and an arrangement direction of the fin in the third heat dissipation member are different from each other.
9. The imaging apparatus according to claim 8, wherein the first heat dissipation member and the third heat dissipation member are disposed in a same space.
10. The imaging apparatus according to claim 1, comprising: an imaging unit that includes an imaging element and a third electronic member; and a fourth heat dissipation member that includes a heat dissipation fin for dissipating heat of the third electronic member, wherein the fourth heat dissipation member includes a first heat dissipation fin and a second heat dissipation fin, and at least a part of the air cooling device is provided between the first heat dissipation fin and the second heat dissipation fin.
11. The imaging apparatus according to claim 1, comprising: a second electronic member that is different from the recording unit; a third heat dissipation member that is provided at an upstream side of the air flow with respect to the air cooling device and that includes a heat dissipation fin that dissipates heat of the second electronic member; an imaging unit that includes an imaging element and a third electronic member; and a fourth heat dissipation member that includes a heat dissipation fin for dissipating heat of the third electronic member, wherein an arrangement interval of the fin in the first heat dissipation member, an arrangement interval of the fin in the third heat dissipation

member, and an arrangement interval of the fin in the fourth heat dissipation member are different from each other.

12. The imaging apparatus according to claim 1, wherein the air cooling device is an air cooling fan.

13. The imaging apparatus according to claim 1, wherein the second heat dissipation member is a metal sheet or a graphite sheet.

14. The imaging apparatus according to claim 1, wherein the recording unit includes a medium accommodation portion in which the recording medium is to be accommodated and that has a lid portion, the lid portion includes a fixing member that dissipates heat of the medium accommodation portion, and the second heat dissipation member is connected to the fixing member.

15. The imaging apparatus according to claim 14, wherein the second heat dissipation member is connected to the first heat dissipation member.

16. An imaging apparatus comprising: a housing that includes a first intake portion, a second intake portion, and an exhaust portion; an air cooling device that is provided inside the housing with an interval from the exhaust portion and that is configured to generate an air flow from the first intake portion and the second intake portion toward the exhaust portion; a rectifying member that is provided at an upstream side of the air flow with respect to the air cooling device and that rectifies intake air from the first intake portion and the second intake portion; a recording unit in which a recording medium that is to record imaging data is to be inserted into and removed from; and a second heat dissipation member that is a heat dissipation sheet that dissipates heat of the recording unit.
